ISPOR EUROPE 2026 · VIENNA

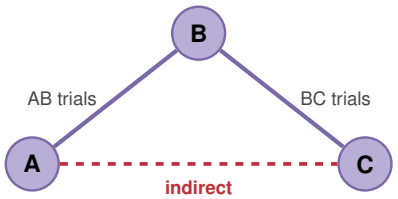
Bridging disconnected networks and single-arm studies with multilevel network meta-regression

ML-NMR-B

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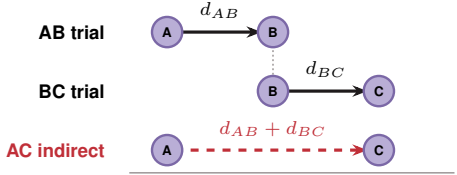
Network meta-analysis

INDIRECT COMPARISON VIA B



$$d_{AC} = d_{AB} + d_{BC}$$

CONSTANCY OF RELATIVE EFFECTS

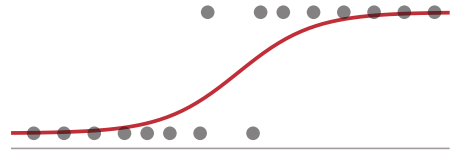


The effect of A vs B is the same in every trial, whichever trial it was measured in.

That's what lets us add d_{AB} and d_{BC}. It holds if **effect modifiers** are balanced across trials.

ML-NMR: one model for IPD and aggregate data

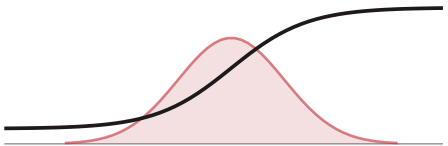
STUDIES WITH IPD: FIT TO INDIVIDUALS



Each dot is one person: responded (top) or not (bottom), by covariate value

$$\eta_{jk}(\boldsymbol{x}) = \mu_j + \boldsymbol{x}^T(\boldsymbol{\beta}_1 + \boldsymbol{\beta}_{2k}) + \gamma_k$$

AGGREGATE DATA: INTEGRATE OVER $f(x)$



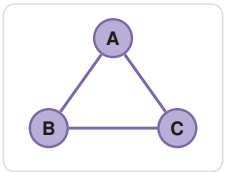
Shaded: the study's covariate distribution, from its reported summaries

$$\theta_{\bullet jk} = \int g^{-1}\{\eta_{jk}(\boldsymbol{x})\}\, f_{jk}(\boldsymbol{x})\, d\boldsymbol{x}$$

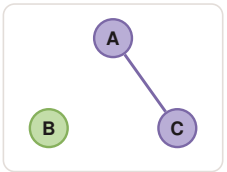
Any network size, any target population. But only for connected networks.

THE PROBLEM

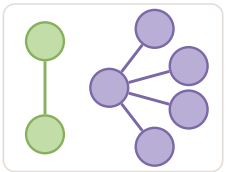
Disconnected networks and single-arm studies



Connected network



Single-arm study



Disconnected networks

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NICE appraisals used population adjustment

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of those were unanchored and used MAIC

10/18

involved more than two treatments

Limitations of MAIC and STC

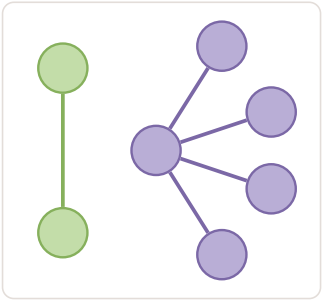
- ✕ One pair of studies at a time
- ✕ Estimates only in the aggregate study's population
- ✕ No way to assess whether the key assumption is plausible

Comparing treatments across a disconnected network assumes:

Conditional constancy of absolute effects

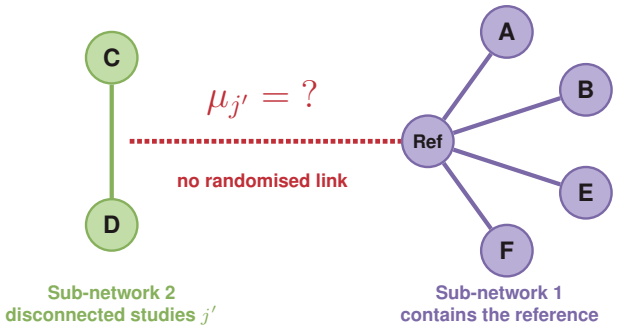
Once covariates are adjusted for, absolute outcomes are exchangeable across studies.

Far stronger than the constancy of relative effects behind a connected NMA, because prognostic imbalance no longer cancels out.



Disconnected networks

Bridge the gap through the study baselines

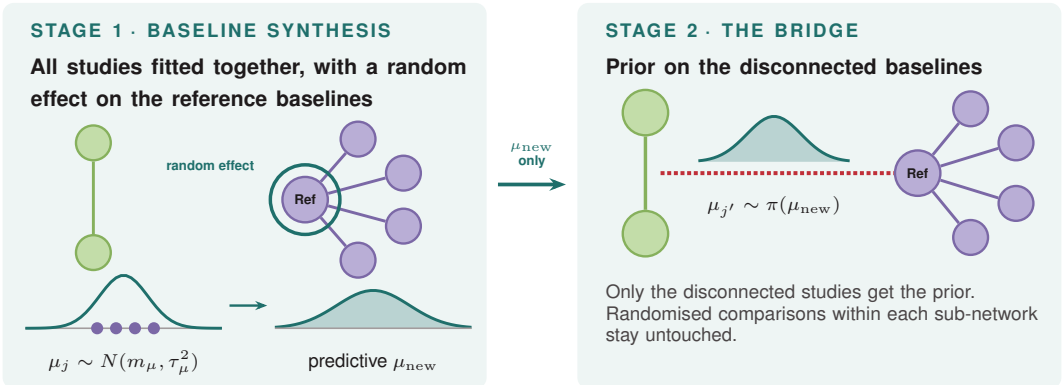


Relative effects within sub-network 2 still come from its own trials (if it has randomised comparisons). Covariate effects are shared across the network.

Missing: $\mu_{j'}$, the outcome on the reference treatment in the disconnected studies.

$$\eta_{j'k}(\boldsymbol{x}) = \mu_{j'} + \boldsymbol{x}^T(\boldsymbol{\beta}_1 + \boldsymbol{\beta}_{2k}) + \gamma_k$$

The random baseline bridge, in two stages



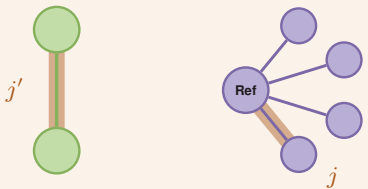
Assumes the adjusted baselines of all bridged studies are exchangeable.

BRIDGE 2

The fixed baseline bridge: merge an anchor pair

STEP 1 · PICK AN ANCHOR PAIR

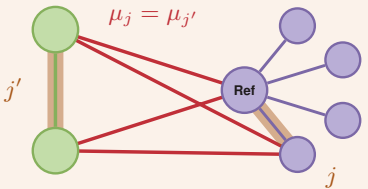
One study from each sub-network



Chosen as the most similar pair of populations (next slide). Can be an internal trial or an external single-arm study.

STEP 2 · SHARE ONE BASELINE

The pair becomes one four-arm study

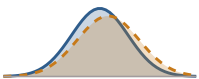


Needed when only one study informs the reference, so τ_μ can't be estimated. No extra uncertainty for unobserved differences.

Assumes the pair's adjusted baselines are identical.

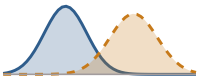
How similar are two populations?

Pool IPD and pseudo-IPD → Model study membership → Weight = P(other study) → **Overlap ESS%**



High overlap

Hard to tell who came from which study. Weights near 0.5, high overlap ESS%.



Low overlap

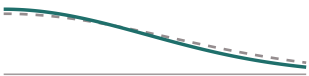
Most people easy to classify. Weights near 0, low overlap ESS%.

Illustrative covariate distributions: solid blue study A, dashed orange study B

Why overlap? Similar on observed covariates ⇒ more likely similar on unobserved ones, with clinical input.

But no guarantee: it only sees observed covariates, so it shows extrapolation, not bias.

τ_μ: do the adjusted baselines still vary?



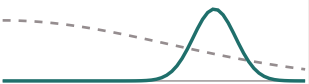
Hugs the prior

Too few studies to tell. Consider an informative prior, or a fixed bridge.



Small and precise

Little unexplained variation among the reference studies.



Large and precise

Adjusted baselines genuinely differ: unobserved factors that may violate the assumption.

Dashed grey: prior for τ_μ · solid teal: posterior from Stage 1

Our best measure of unobserved differences. Studies in an NMA already share a PICO and a clinical rationale for being combined, so the variation among the reference studies is a reasonable guide for the bridged ones.

Checking the model and the bridging choices

PREDICTIVE PERFORMANCE

Can the outcome model predict well?

- Leave-one-out cross-validation in the IPD studies
- Checks the covariates are informative about outcomes, compared with an unadjusted model
- Checks the model is well estimated, so we can predict into the disconnected studies and target populations
- Not a direct test of the bridge: it's within-study and individual-level

SENSITIVITY ANALYSIS ACROSS BRIDGES

Does the answer depend on the bridge?

Fit a set of plausible bridging models, not just one:

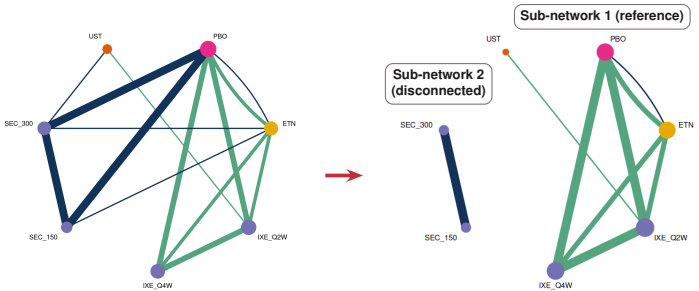
1. Random bridge: prior on one study vs several
2. Fixed bridges through different anchor pairs
3. Random vs fixed on the same studies

Every model still rests on the same assumption, so this **doesn't test it**. It shows how far conclusions depend on the bridging choice: report changes rather than pick one model.

Neither tests the assumption: with τ_μ and overlap, they help judge it.

CASE STUDY

Psoriasis: break a network, rebuild it



- 9 RCTs: 4 IPD, 5 AgD
- 6 active treatments and placebo
- PASI 75/90/100 at 12 weeks
- Secukinumab cut off from PBO
- Each bridge vs the full network

Fully connected (benchmark)

Random-1P · prior on ERASURE

Random-3P · prior on all three

Artificially disconnected

Fixed · UNC-3 + ERA (97%)

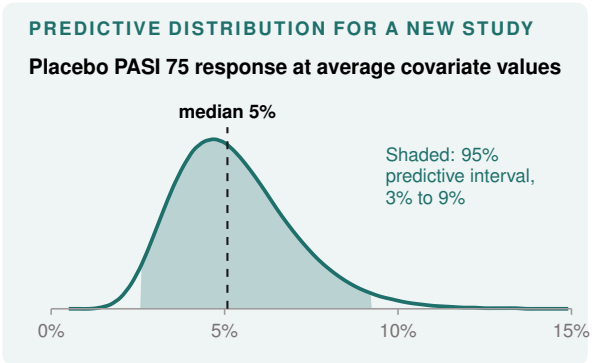
Fixed · JUNC + IXOR (74%)

Single-arm · PROSPECT (89%)

Single-arm · Chiricozzi (77%)

Data: Phillippo et al. 2023; IPD simulated to resemble the trials · %: overlap ESS of the anchor pair

Stage 1 in practice: the psoriasis baseline synthesis



Posterior summaries (probit scale)

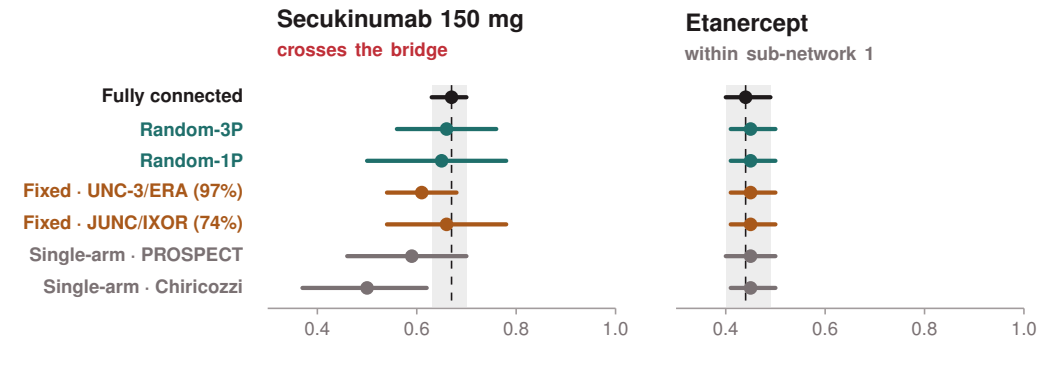
Mean baseline m_μ	-1.64 $(-1.81, -1.46)$
Between-study SD τ_μ	0.115 $(0.029, 0.323)$
New study μ_{new}	-1.64 $(-1.95, -1.31)$

4 placebo-controlled studies in sub-network 1 (FIXTURE, UNCOVER-1/2/3).

τ_μ is **small and precisely estimated**: the data have pulled it well away from its vague prior.

This distribution becomes the prior for the disconnected baselines.

Only bridged comparisons depend on the bridge



Etanercept: every bridge matches the full network

Random-3P sits closest; 1P is wider

74% pair as close as 97%: overlap $\neq$ bias

Chiricozzi drifts low: likely unobserved differences

ML-NMR posterior mean and 95% CrI · dashed line and band: fully connected estimate

TAKE-HOMES

What ML-NMR-B gives you

1 **Population adjustment across disconnected networks**

Builds on baseline-exchangeability models (Béliveau 2017; Thom 2022), adding IPD-level adjustment and a two-stage fit.

2 **Allows for unobserved differences**

The random baseline bridge carries residual between-study variation through as uncertainty.

3 **Makes the key assumption judgeable**

τ_μ, overlap ESS and predictive checks, plus sensitivity analyses across bridges.

Next: a simulation study will test the bias and coverage of each bridging strategy, including when the bridging assumptions fail.

Implemented in multinma: coming to CRAN shortly

THANK YOU

Questions?

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SOFTWARE

multinma · on CRAN shortly

dmphillippo.github.io/multinma

WITH

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